

ETHERNET

Random Access Protocols

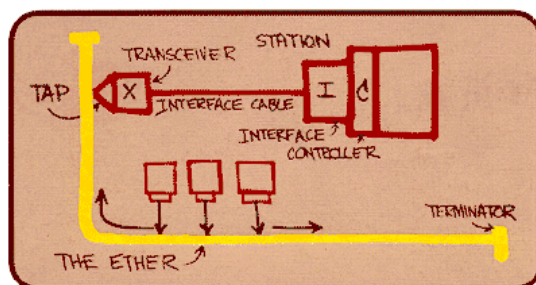
- When node has packet to send
 - Transmit at full channel data rate R .
 - No a priori coordination among nodes
- Two or more transmitting nodes → “collision”
- Random access MAC protocol specifies:
 - How to detect collisions
 - How to recover from collisions
- Examples
 - CSMA/CD (Ethernet: IEEE 802.3)
 - CSMA/CA (Wireless: IEEE 802.11)

Key Ideas of Random Access

- Carrier sense
 - *Listen before speaking, and don't interrupt*
 - Checking if someone else is already sending data
 - ... and waiting till the other node is done
- Collision detection
 - *If someone else starts talking at the same time, stop*
 - Realizing when two nodes are transmitting at once
 - ...by detecting that the data on the wire is garbled
- Randomness
 - *Don't start talking again right away*
 - Waiting for a random time before trying again

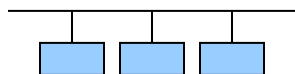
Ethernet

- Dominant wired LAN technology
- First widely used LAN technology
- Simpler, cheaper than token LANs and ATM
- Kept up with speed race: 10 Mbps – 10 Gbps



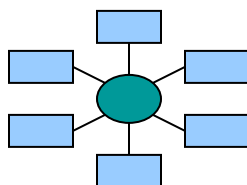
Metcalfe's
Ethernet
sketch

Ethernet Topologies



Bus Topology: Shared

All nodes connected
to a wire



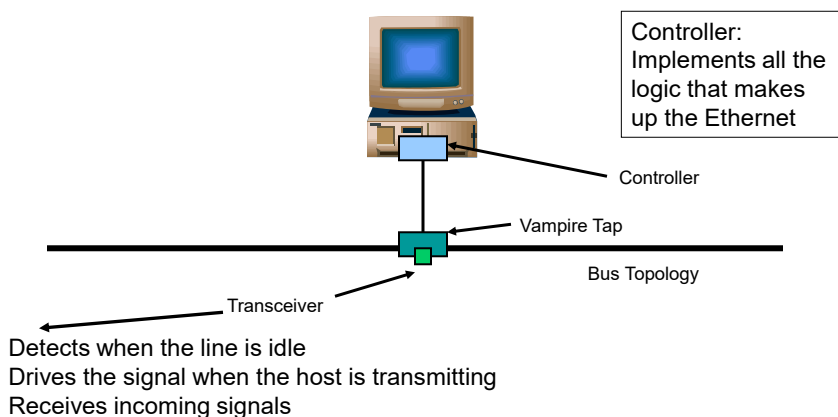
Star Topology:

All nodes connected to a
central repeater

Ethernet Connectivity

10Base5 – ThickNet

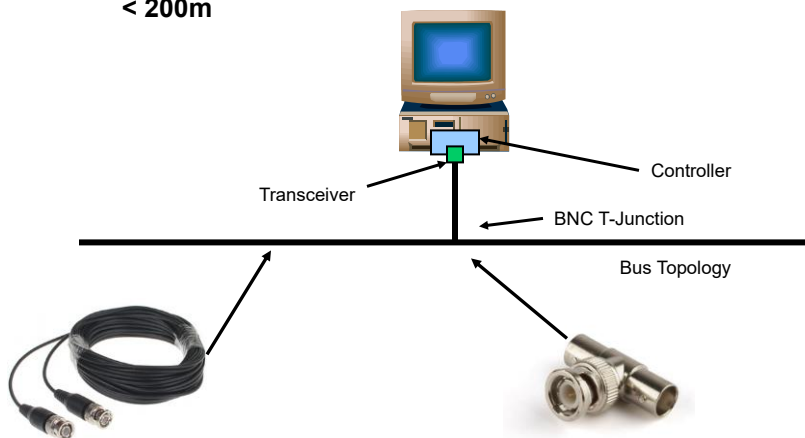
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Ethernet Connectivity

10Base2 – ThinNet

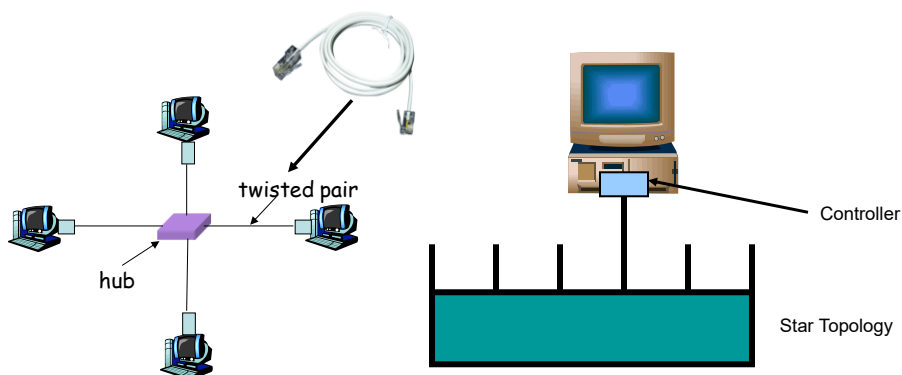
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Ethernet Connectivity

10BaseT

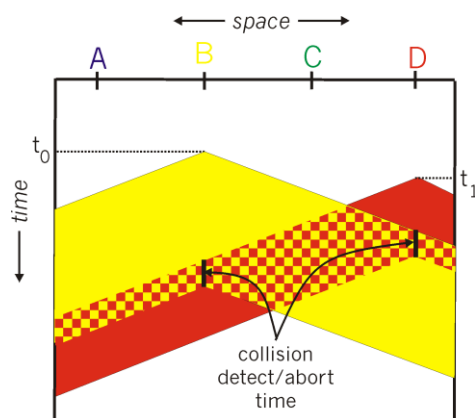
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CSMA/CD (Collision Detection)

- Carrier Sense Multiple Access, CSMA: listen before transmit
 - If channel sensed idle: transmit entire frame
 - If channel sensed busy, defer transmission
- CD, Collision Detection :
 - Collisions detected *within short time*
 - Colliding transmissions *aborted*, reducing wastage
 - Easy in wired LANs: measure signal strengths, compare transmitted, received signals
 - Difficult in wireless LANs: receiver shut off while transmitting
- Human analogy: the polite conversationalist

CSMA/CD Collision Detection



Ethernet MAC Protocol, CSMA/CD

- Carrier sense: wait for link to be idle
 - Channel idle: start transmitting
 - must wait $9.6\mu s$ between back-to-back frames
 - Channel busy: wait until idle and transmit immediately
- Collision detection: listen while transmitting
 - No collision: transmission is complete
 - Collision: abort transmission, and send a 32-bit “jamming sequence”

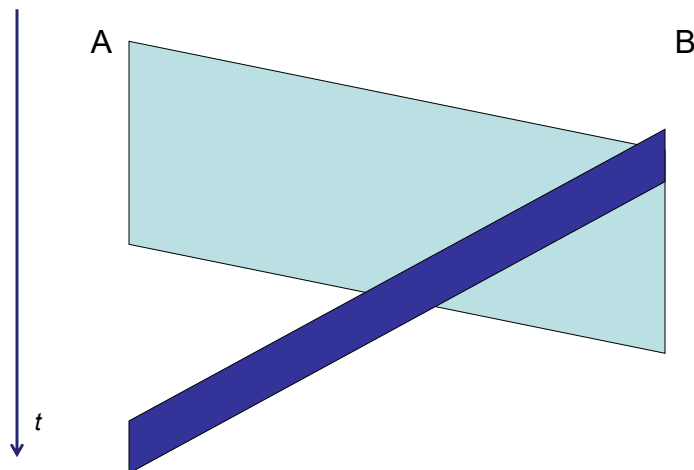
Ethernet MAC Protocol, CSMA/CD(cont.)

- Random access: exponential back-off
 - After collision, wait a random time before trying again
 - After m^{th} collision, choose K randomly from $\{0, \dots, 2^m - 1\}$
 - ... and wait for $K \cdot 512$ bits transmission time before trying again
 - $K \cdot 0.1\mu s$ on 10Mbps Ethernet

Delay Times After Collision

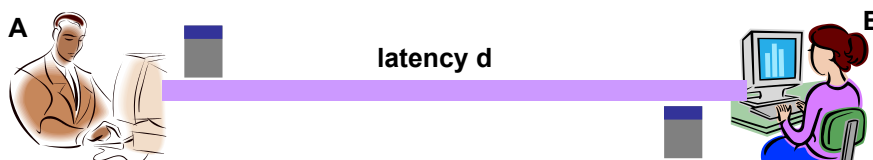
- Delay and try again
 - 1st time: 0 or $0.1\mu\text{s}$
 - 2nd time: 0, 0.1, or $0.2\mu\text{s}$
 - 3rd time: 0, 0.1, 0.2, or $0.3\mu\text{s}$
 - After n collisions,
 - wait* time: $k \times 0.1\mu\text{s}$,
 - for randomly selected $k=0 \dots 2^n - 1$
 - give up after several tries (up to 16)
 - exponential backoff

Collisions



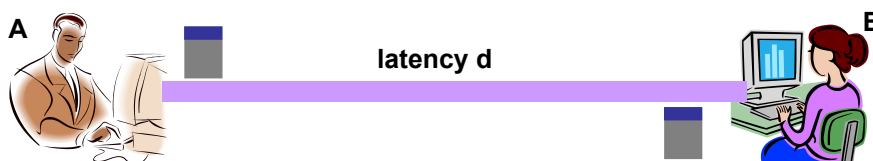
How can we ensure that A knows about the collision?

Limitations on Ethernet Length



- Latency depends on physical length of link
 - Time to propagate a packet from one end to the other
- Suppose A sends a packet at time t
 - And B sees an idle line at a time just before $t+d$
 - ... so B happily starts transmitting a packet
- B detects a collision, and sends jamming signal
 - But A doesn't see collision till $t+2d$

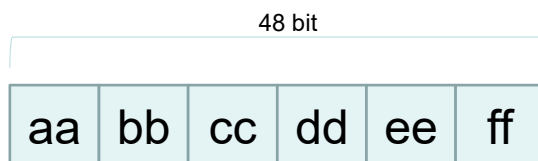
Limitations on Ethernet Length



- A needs to wait for time $2d$ to detect collision
 - So, A should keep transmitting during this period
 - ... and keep an eye out for a possible collision
- Imposes restrictions on Ethernet
 - Maximum length of the wire: 2500 meters
 - Minimum length of the packet: 512 bits (64 bytes)
- 802.3
 - $2d$ is bounded to $51.2\mu\text{s}$
 - At 10Mbps, and during $51.2\mu\text{s}$, 512bit are transmitted (64B)
 - Packet length $\geq 64\text{B}$

Ethernet MAC Address

- 48 bit Ethernet Addresses



- Typically written in six octets in hex
Example: **00-15-C5-48-04-A9**
- $2^{48} > 281 \times 10^{12}$ Addresses available

MAC Addresses

- Organizationally unique identifier
 - Assigned by IEEE Registration Authority
 - Determines the first 24 bits of the address
 - Example: **00-15-C5** corresponds to “Dell Inc”
- Remainder of the MAC address
 - Allocated by the manufacturer
 - Example: **48-04-A9**

Special Addresses

- Broadcast

FF-FF-FF-FF-FF-FF

Used to send frames to all hosts on an Ethernet

- Multicast

Xxxx1-XX-XX-XX-XX-XX,

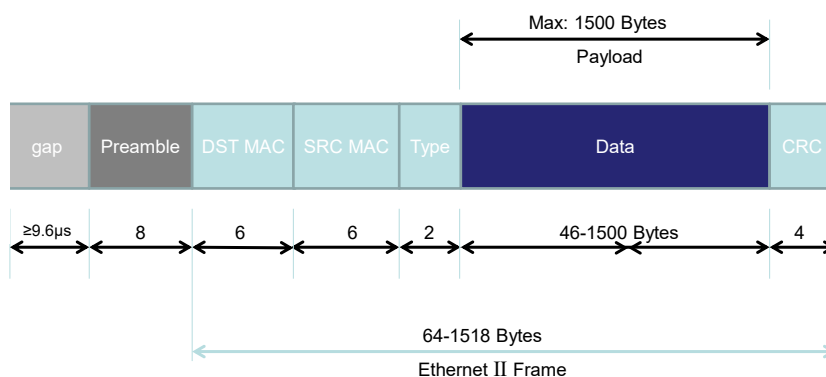
X = any nibble

x = any bit

Adaptors can be programmed to accept some set of multicast addresses

Ethernet Frame

Sending adapter encapsulates payload in **Ethernet frame**



Ethernet Frame Structure

- Sending adapter encapsulates IP datagram (or other network layer protocol packet) in **Ethernet frame**

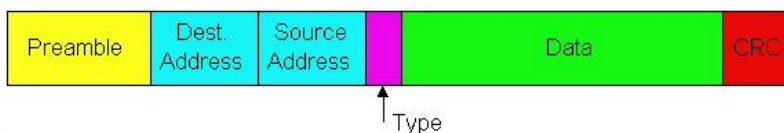


Preamble: 8 bytes

- 7 bytes with pattern 10101010 followed by one byte with pattern 10101011
- Used to synchronize receiver, sender clock rates (Manchester encoding)

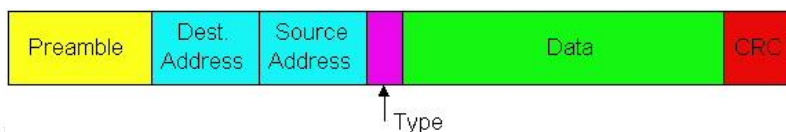
Ethernet Frame Structure (2)

- **Address:** 6 bytes
 - unique, 48-bit unicast address assigned to each adapter ($2^{48} > 281 \times 10^{12}$);
 - human readable, burned in ROM (example: 8:0:2b:e4:b1:2 (00001000 00000000 00101011 11100100 10110001 00000010))
 - broadcast: all 1s
 - multicast: first bit is 1
 - used to send messages to some subset of the hosts on an Ethernet (e.g. all file servers)
 - adaptor programmed to accept some set of multicast addresses



Ethernet Frame Structure (3)

- Adaptor receives all frames and accepts:
 - Frames addressed to its address;
 - Frames addressed to the broadcast address;
 - Frames addressed to a multicast address, if it has been instructed to listen to that address;
 - All frames, if it has been placed in promiscuous mode.
- **Type:** 2B; indicates the higher layer protocol (mostly IP but others may be supported such as Novell IPX and AppleTalk)
- **CRC:** 4B; checked at receiver, if error then frame is discarded



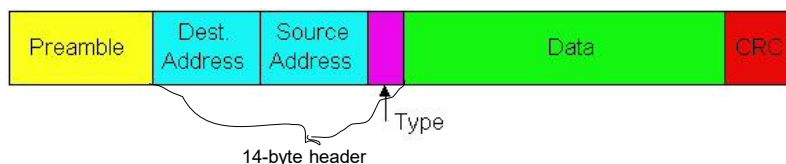
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Ethernet Frame Structure (4)

- **Frame Data: Up to 1500 data bytes**
 - Minimally 46 bytes of data
 - Minimal frame = 46 (data) + 14 (header) + 4 (CRC) = 64 bytes
 - The reason for a minimum frame size is that the frame must be long enough to detect a collision



isep Instituto Superior de Engenharia do Porto

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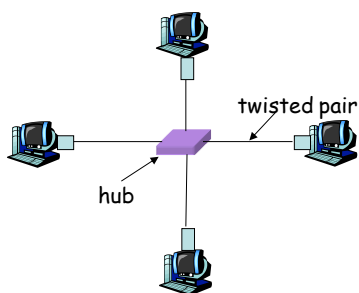
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Unreliable, Connectionless Service

- Connectionless
 - No handshaking between sending and receiving adapter.
- Unreliable
 - Receiving adapter doesn't send ACKs
 - Packets passed to network layer can have gaps
 - Gaps will be filled if application is using TCP
 - Otherwise, the application will see the gaps

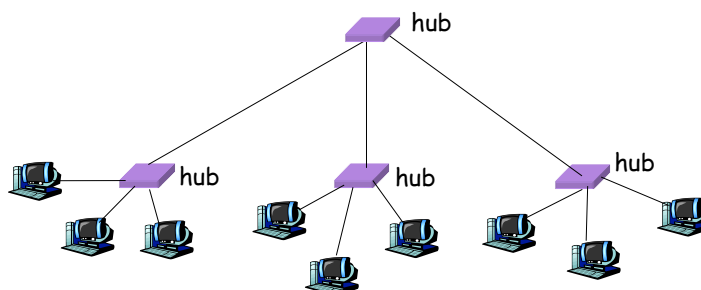
Hubs: Physical-Layer Repeaters

- Hubs are physical-layer repeaters
 - Bits coming from one link go out all other links
 - At the same rate, with no frame buffering
 - No CSMA/CD at hub: adapters detect collisions



Interconnecting with Hubs

- Backbone hub interconnects LAN segments
- All packets seen everywhere, forming one large collision domain
- Can't interconnect Ethernets of different speeds

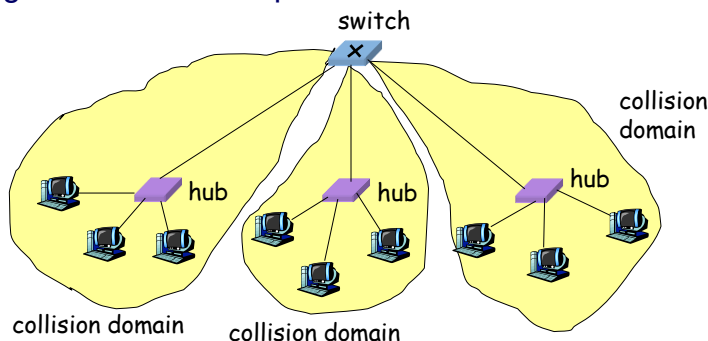


Switch

- Link layer device
 - Stores and forwards Ethernet frames
 - Examines frame header and selectively forwards frame based on MAC destination address
 - When frame is to be forwarded on segment, uses CSMA/CD to access segment
- Transparent
 - Hosts are unaware of presence of switches
- Plug-and-play, self-learning
 - Switches do not need to be configured

Switch: Traffic Isolation

- Switch breaks subnet into LAN segments
- Switch filters packets
 - Same-LAN-segment frames not usually forwarded onto other LAN segments
 - Segments become separate collision domains

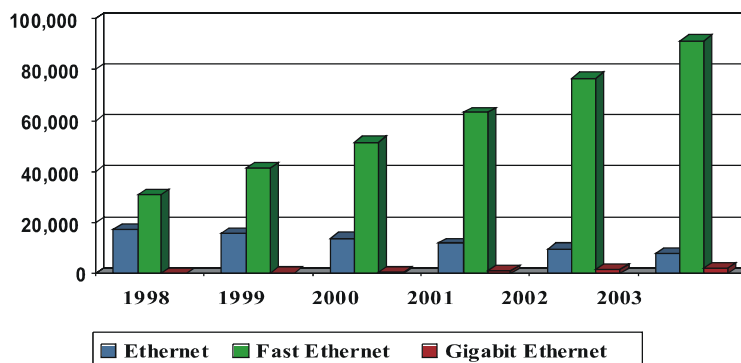


Scalability Challenges

- MAC addresses are flat
 - Multiple hosts on the same network
 - No relationship between MAC addresses
- Data plane
 - Forwarding based on MAC address
 - Table size? Look-up overhead?
- Control plane
 - Determining where the host is located
 - Keeping the information up-to-date

WW LAN NIC Market

System Units (K)

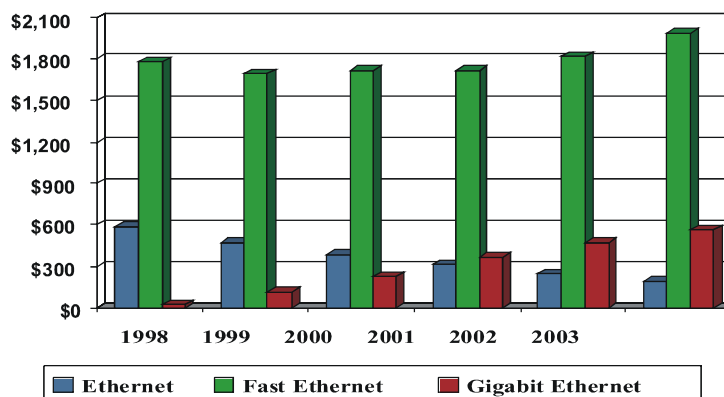


Ethernet: 10 Mbps
Fast Ethernet : 100 Mbps
Gigabit Ethernet : 1 Gbps

Source: Dataquest

WW LAN NIC Market

Revenue (\$M)

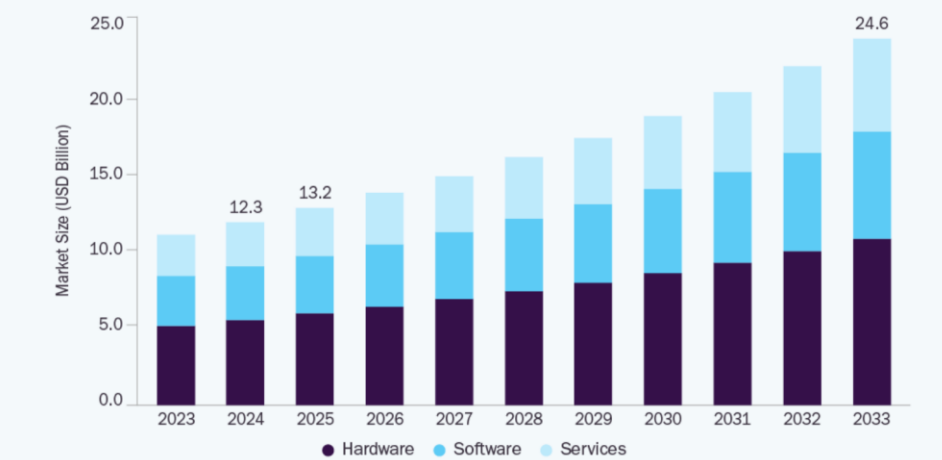


Ethernet: 10 Mbps
Fast Ethernet : 100 Mbps
Gigabit Ethernet : 1 Gbps

Source: Dataquest

Industrial Ethernet Market

Size, by Type, 2023 - 2033 (USD Billion)



Benefits of Ethernet

- Easy to administer and maintain
- Inexpensive
- Increasingly higher speed
- Moved from shared media to switches
 - Change everything except the frame format
 - A good general lesson for evolving the Internet

Ethernet physical layer

Designation	Supported Media	Maximum Segment Length	Transfer Speed	Topology
10Base-5	Coaxial	500m	10Mbps	Bus
10Base-2	ThinCoaxial	185m	10Mbps	Bus
10Base-T	UTP cat3	100m	10Mbps	Star, hub or switch
1Base-5	UTP cat3	100m	1Mbps	Star, hub
10Broad-36	Coaxial	3600m	10Mbps	Bus
10Base-FL	Fiber-optic	2000m	10Mbps	Star
100Base-TX	Cat5 UTP	100m	100Mbps	Star, hub or switch
100Base-FX	Fiber-optic	2000 m	100 Mbps	Star
1000Base-SX	Fiber-optic	260m	1Gbps	Star
1000Base-LX	Fiber-optic	5000 m	1Gbps	Star
1000Base-CX	Twinax	25m	1Gbps	Star
1000Base-T	Category5	100m	1Gbps	Star

Ethernet physical layer

Designation	Supported Media	Block Coding	Line Coding	Topology
10Base-5	Coaxial			Bus
10Base-2	ThinCoaxial			Bus
10Base-T	UTP cat3	Manchester	Differential	Star, hub or switch
1Base-5	UTP cat3			Star, hub
10Broad-36	Coaxial			Bus
10Base-FL	Fiber-optic			Star
100Base-TX	Cat5 UTP	4b/5b	MLT-3	Star, hub or switch
100Base-FX	Fiber-optic	4b/5b	NRZI	Star
1000Base-SX	Fiber-optic	8b/10b	NRZ	Star
1000Base-LX	Fiber-optic	8b/10b	NRZ	Star
1000Base-CX	Twinaxial STP	8b/10b		Star
1000Base-T	Category5		4D-PAM5	Star
10GBase-T	Cat6 UTP	64b/65b + LDPC	DSQ128-PAM16	Star
40GBase-R	Cat8 TP	64b/66b		Star

4B/5B

Block Coding

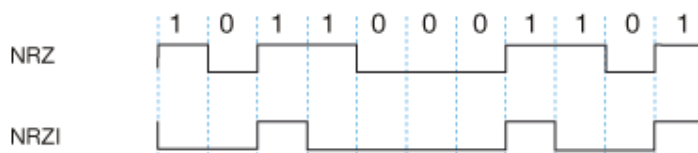
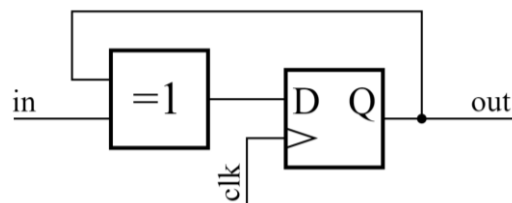
Each Nibble is coded in a 5 bit word.

- Provides transitions for clock retrieval.
- Unused codes can be applied to control.

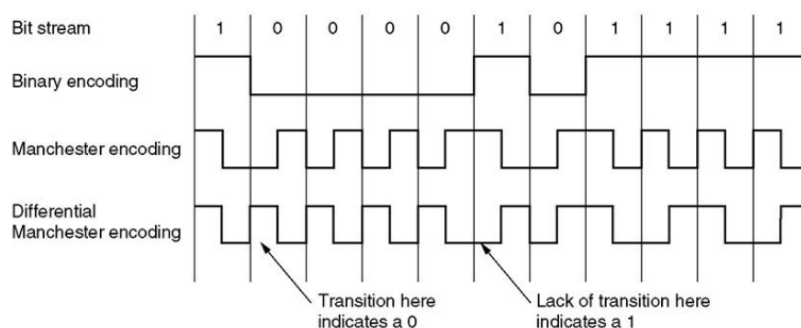
Hex	Bin	4B/5B
0	0000	11110
1	0001	01001
2	0010	10100
3	0011	10101
4	0100	01010
5	0101	01011
6	0110	01110
7	0111	01111
8	1000	10010
9	1001	10011
A	1010	10110
B	1011	10111
C	1100	11010
D	1101	11011
E	1110	11100
F	1111	11101

NRZI

- An output transition is performed whenever input is 1.

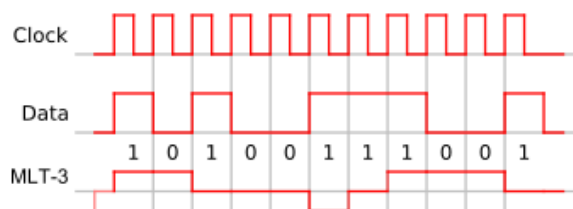


Manchester coding



MLT-3

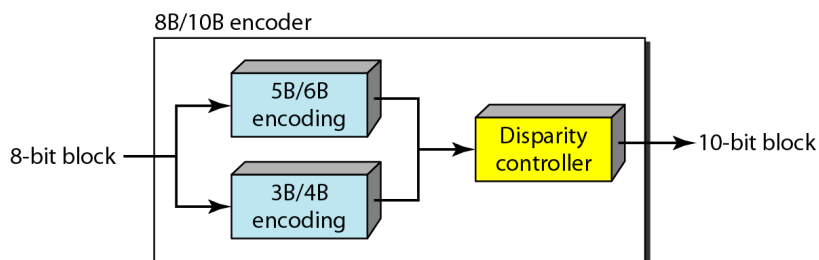
- MLT-3: Multi-Level Transmit, 3 levels
- Output follows pattern:
0, +1, 0, -1, 0, +1, 0,
- An output transition is performed whenever input is 1.



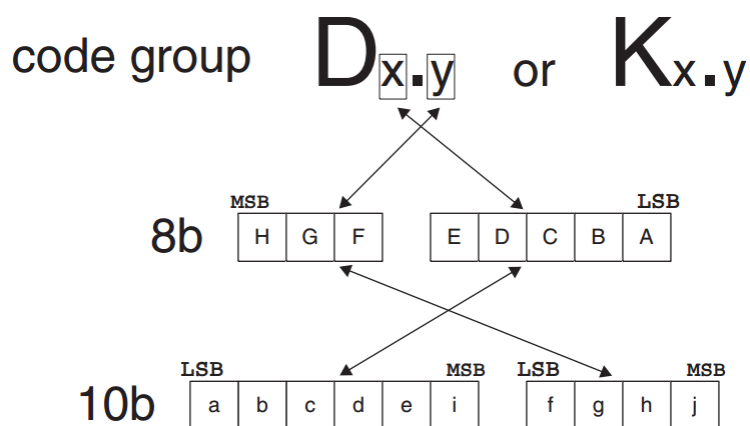
NRZ	0	0	0	1	0	0	1	0	1	0	0
MLT-3	0	0	0	+	+	+	0	0	-	0	+

8B/10B

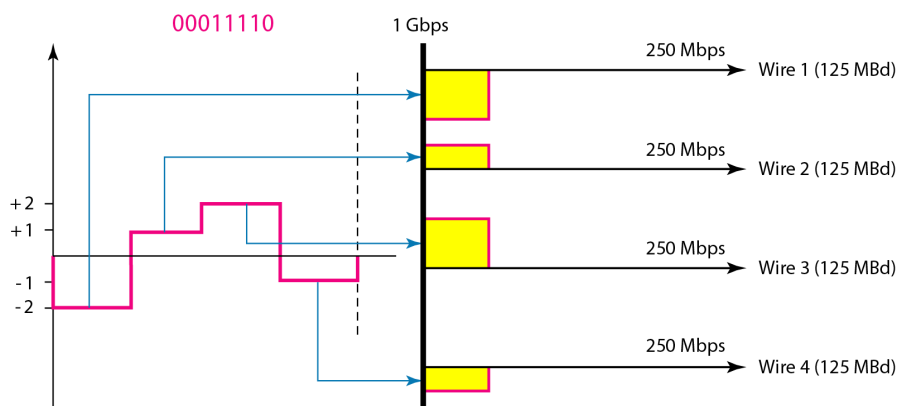
Proposed by Albert X. Widmer and Peter A. Franaszek of IBM Corporation in 1983



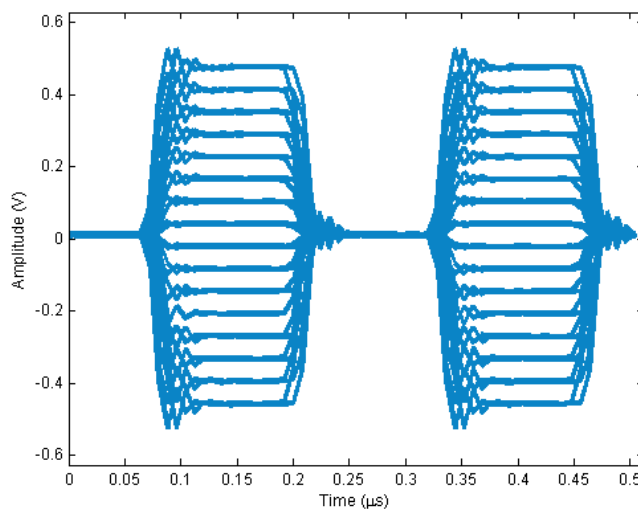
8b/10b



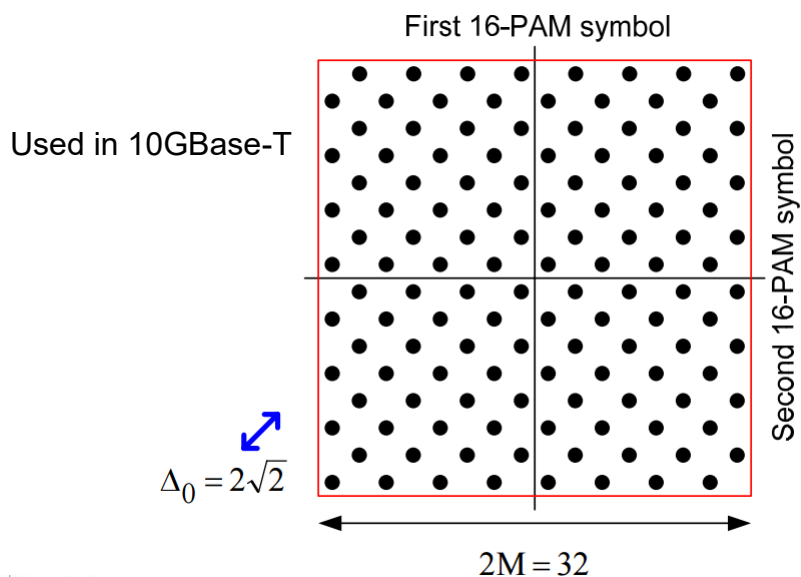
4D-PAM5



PAM16



128-DSQ



Ethernet line levels

